satisfied [325] is

$$b_i a_{ij} + b_j a_{ji} = b_i b_j, \qquad i = 1, \ldots, s, \qquad j = 1, \ldots s.$$

The implicitness reduces the usefulness of Runge-Kutta methods for applications like molecular dynamics, because the complexity and cost of the molecular force laws usually means that implicit methods are inefficient (this is certainly not prima facie obvious, but the issue has been examined by several researchers without much positive result; see, e.g. [308]).

Example 2.7 The Gauss-Legendre family of Runge-Kutta (GLRK) methods correspond to approximating the vector field at the Gauss points, i.e. the zeros of the orthogonal polynomials that arise in Gaussian quadrature. As these points are symmetrically distributed the GLRK schemes are symmetric, hence have even order. The simplest such method is the *implicit midpoint* rule:

$$\boldsymbol{Z} = z + h\boldsymbol{F}_1, \qquad \boldsymbol{F}_1 = \boldsymbol{f}(z + \frac{h}{2}\boldsymbol{F}_1),$$

which has order 2. The 4th order method ($s = 2$) has coefficients

$$b_1 = b_2 = \frac{1}{2}, \qquad A = (a_{ij}) = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \end{bmatrix}.$$

2.5.2 Partitioned Runge-Kutta Methods

It can be seen that Runge-Kutta methods treat all components of the differential equation identically. On the other hand, in molecular dynamics the equations of motion often have a special structure: for example the differential equation system is typically linear in $\boldsymbol{p}$, and, moreover, the equations have a special coupling structure so that the differential equation for $\boldsymbol{q}$ depends only on $\boldsymbol{p}$ and that for $\boldsymbol{p}$ depends only on $\boldsymbol{q}$. So-called partitioned Runge-Kutta methods allow us to exploit this structure. As an illustration, consider the method:

$$\hat{\boldsymbol{P}} = \boldsymbol{p} - \frac{h}{2}\nabla_{\boldsymbol{q}} H(\boldsymbol{q}, \hat{\boldsymbol{P}}), \tag{2.26}$$

$$\boldsymbol{Q} = \boldsymbol{q} + \frac{h}{2}(\nabla_{\boldsymbol{p}} H(\boldsymbol{q}, \hat{\boldsymbol{P}}) + \nabla_{\boldsymbol{p}} H(\boldsymbol{Q}, \hat{\boldsymbol{P}})), \tag{2.27}$$

$$\boldsymbol{P} = \hat{\boldsymbol{P}} - \frac{h}{2}\nabla_{\boldsymbol{q}} H(\boldsymbol{Q}, \hat{\boldsymbol{P}}). \tag{2.28}$$